

• Notes for Lectures 8-9 (Part B of the course)

Let T be a translation-invariant CZO that is, there exists a $u \in \mathcal{S}'(\mathbb{R}^d)$ such that $T(f) := u * f$, $f \in \mathcal{S}(\mathbb{R}^d)$, and:

(I) • $\hat{u}$ agrees with an L^∞ -function,

(II) • u agrees with a function $K \in L^1_{loc}(\mathbb{R}^d \setminus \{0\})$ on $\mathbb{R}^d \setminus \{0\}$ that satisfies

$$A = \sup_{y \in \mathbb{R}^d} \int_{|x| > 2|y|} |K(x-y) - K(x)| dx < \infty.$$

As we have seen, T can be extended as an L^p -bounded operator for all $p \in (1, \infty)$. However, there exist CZOs that are not bounded on $L^1(\mathbb{R}^d)$; for instance, the Hilbert transform is not bounded on $L^1(\mathbb{R})$. In general, for an arbitrary function $f \in L^1(\mathbb{R}^d)$ the best that one can say is that $T(f) \in L^{1,\infty}(\mathbb{R}^d)$ i.e. $\sup_{a>0} (a \cdot |\{ |T(f)(x)| > a \}|) < \infty$.

Our goal in this lecture is to determine a subspace X of $L^1(\mathbb{R}^d)$ such that $T(f) \in L^1(\mathbb{R}^d)$ for all $f \in X$ and X is a certain sense best possible. Towards this aim, let T be a given translation-invariant CZO. Recall that in the proof of the fact that T is of weak type $(1,1)$, we considered a function $f \in \mathcal{S}(\mathbb{R}^d)$ and an $a > 0$ and then decomposed f as $f = g + b$ with g and b arising from the Calderón-Zygmund decomposition of f at 'height' a . To deal with the 'good part' g , we only used the L^2 -boundedness of T (which is a direct consequence of (I) thanks to Plancherel's theorem). To handle the contribution of the 'bad part' b , we split

$$\{x \in \mathbb{R}^d : |T(b)(x)| > \alpha/2\} = \underbrace{\left\{x \in \bigcup_{Q \in \mathcal{Q}} Q^* : |T(b)(x)| > \alpha/2\right\}}_{B_1} \cup \underbrace{\left\{x \in \mathbb{R}^d \setminus \bigcup_{Q \in \mathcal{Q}} Q^* : |T(b)(x)| > \alpha/2\right\}}_{B_2},$$

where Q^* is an appropriate enlargement of Q , $Q \in \mathcal{Q}$.
 - To handle B_1 , we simply used properties of $Q \in \mathcal{Q}$ and showed that $|B_1| \leq \left| \bigcup_{Q \in \mathcal{Q}} Q^* \right| \leq C_\alpha \|f\|_{L^1(\mathbb{R}^d)}$.

- The fact that $|B_2| \leq C'_\alpha \|f\|_{L^1(\mathbb{R}^d)}$ was a consequence of the fact that

$$(*) \quad \int_{\mathbb{R}^d \setminus Q^*} |T(b_Q)(x)| dx \leq C'' \cdot \int_Q |b_Q(x)| dx$$

where $b_Q := b \cdot \chi_Q$.

The proof of $(*)$ was obtained by using the fact that:

fact 1: if $h \in L^2(\mathbb{R}^d)$ and $\text{supp}(h) \subseteq F$, $F \subseteq \mathbb{R}^d$ compact, then $T(h)(x) = \int_{\text{supp}(h)} K(x-y) h(y) dy$ for a.e. $x \in \mathbb{R}^d \setminus (\gamma F)$, $\gamma > 1$.

combined with (II) and the fact that

$$(1) \quad \int_Q b_Q(x) dx = 0.$$

Hence, if we want to come up with a suitable candidate $X \subseteq L^1(\mathbb{R}^d)$ such that $T(f) \in L^1(\mathbb{R}^d)$ for all $f \in X$, then a possible approach is to consider first functions that behave like the functions b_Q considered above. More specifically, start with a function $a_Q \in L^1(\mathbb{R}^d)$ associated to a cube $Q \subseteq \mathbb{R}^d$ so that:

$$(i') \quad \text{supp}(a_Q) \subseteq Q,$$

$$(ii') \quad \int_Q a_Q(y) dy = 0.$$

By arguing as in the proof of (*), our assumptions (i'), (ii') imply that

$$(a') \quad \int_{\mathbb{R}^d \setminus Q^*} |T(a_Q)(x)| dx \leq C \int_Q |a_Q(x)| dx,$$

where Q^* is concentric to Q with side-length $\ell(Q^*) = 2d^{1/2} \ell(Q)$ and C is a constant that depends on d, A , and $\|\hat{u}\|_{L^\infty(\mathbb{R}^d)}$ and not on a_Q .

Remember that our goal is to come up with functions $f \in L^1(\mathbb{R}^d)$ such that $T(f) \in L^1(\mathbb{R}^d)$. In view of (a'), we need to handle the term

$$\int_{Q^*} |T(a_Q)(x)| dx.$$

To this end, observe that the Cauchy-Schwarz inequality combined with the L^2 -boundedness of T yields that

$$\begin{aligned} \int_{Q^*} |T(a_Q)(x)| dx &\stackrel{C-5}{\leq} |Q^*|^{1/2} \left(\int_{\mathbb{R}^d} |T(a_Q)(x)|^2 dx \right)^{1/2} \\ &\stackrel{\|T\|_{L^2 \rightarrow L^2} = \|\hat{u}\|_{L^\infty(\mathbb{R}^d)}}{\leq} \|\hat{u}\|_{L^\infty(\mathbb{R}^d)} \cdot |Q^*|^{1/2} \left(\int_Q |a_Q(x)|^2 dx \right)^{1/2} \\ &= 2^d d^{d/2} \cdot \|\hat{u}\|_{L^\infty(\mathbb{R}^d)} \cdot |Q|^{1/2} \left(\int_Q |a_Q(x)|^2 dx \right)^{1/2} \end{aligned}$$

Hence, if we impose the extra assumption that

$$(iii') \quad \left(\int_Q |a_Q(x)|^2 dx \right)^{1/2} \leq |Q|^{-1/2},$$

we deduce that

$$(b') \quad \int_{Q^*} |T(a_Q)(x)| dx \leq 2^d d^{d/2} \cdot \|\hat{u}\|_{L^\infty(\mathbb{R}^d)}.$$

So, if a_Q satisfies (i'), (ii'), (iii') then $T(a_Q) \in L^1(\mathbb{R}^d)$ in view of (a'), (b').

The observations made above motivate the following definition

Definition 1. Let $Q \subseteq \mathbb{R}^d$ be a cube. A function a_Q is said to be an **atom** associated to Q if:

$$(i') \quad \text{supp}(a_Q) \subseteq Q,$$

$$(ii') \quad \int_Q a_Q(x) dx = 0,$$

$$(iii') \quad \int_Q |a_Q(x)|^2 dx \leq |Q|^{-1}.$$

Exercise 2. Let T be a CZO as above.

Prove that there exists a constant $C > 0$, depending only on T and d , such that

$$\int_{\mathbb{R}^d} |T(a_Q)(x)| dx \leq C$$

for all atoms a_Q .

Definition 3. Define X to be the class of all $f \in L^1(\mathbb{R}^d)$ such that there exists a sequence of complex numbers $\{\mu_k\}_{k \in \mathbb{N}} \in \ell^1(\mathbb{N})$ and a sequence of atoms $\{a_{Q_k}\}_{k \in \mathbb{N}}$ such that $f = \sum_{k \in \mathbb{N}} \mu_k a_{Q_k}$ (in the L^1 -norm).

$$\text{Set } \|f\|_X := \inf \left\{ \sum_{k \in \mathbb{N}} |\mu_k| : f = \sum_{k \in \mathbb{N}} \mu_k a_{Q_k} \right\}.$$

Exercise 4. Show that $(X, \|\cdot\|_X)$ is complete.

Theorem 5. Let T be a translation-invariant CZO. Then $T(f) \in L^1(\mathbb{R}^d)$ for all $f \in X$.

Proof. Let $f \in X$ be given. Let $\{\mu_k\}_{k=1}^{\infty} \in \ell^1(\mathbb{N})$ and let $\{a_{Q_k}\}_{k=1}^{\infty}$ be a sequence of atoms such that

$$f = \sum_{k=1}^{\infty} \mu_k a_{Q_k} \quad \text{in the } L^1\text{-norm.}$$

• As T is of weak-type $(1,1)$, $T(f)$ is well-defined.

• It follows from exercise 2 that $\|T(a_{Q_k})\|_{L^1(\mathbb{R}^d)} \leq C$ for all $k \in \mathbb{N}$ and so,

$$\sum_{k=1}^{\infty} \mu_k T(a_{Q_k})$$

converges in the L^1 -norm to some $g \in L^1(\mathbb{R}^d)$ with

$$(2) \quad \|g\|_{L^1(\mathbb{R}^d)} \leq C \cdot \sum_{k \in \mathbb{N}} |\mu_k|.$$

• We claim that

$$(3) \quad T(f) = g \quad \text{a.e. on } \mathbb{R}^d.$$

Indeed, for any $\varepsilon > 0$, we have

$$|\{x \in \mathbb{R}^d : |T(f)(x) - g(x)| > \varepsilon\}| =$$

$$|\{x \in \mathbb{R}^d : |T(f)(x) - \sum_{k=1}^{\infty} \mu_k T(a_{Q_k})(x)| > \varepsilon\}| \leq$$

$$|\{x \in \mathbb{R}^d : |T(f)(x) - \sum_{k=1}^N \mu_k T(a_{Q_k})(x)| > \varepsilon/2\}| +$$

$$|\{x \in \mathbb{R}^d : |\sum_{k=N}^{\infty} \mu_k T(a_{Q_k})(x)| > \varepsilon/2\}| \leq \overset{\substack{T \text{ is linear and} \\ \text{of weak-type } (1,1)}}{\leq}$$

$$\frac{C'}{\varepsilon} \int_{\mathbb{R}^d} |f(x) - \sum_{k=1}^N \mu_k a_{Q_k}(x)| dx + \frac{2}{\varepsilon} \int_{\mathbb{R}^d} |\sum_{k=N}^{\infty} \mu_k T(a_{Q_k})(x)| dx \leq \overset{\text{Ex. 2}}{}$$

$$2C' \frac{1}{\varepsilon} \|f - \sum_{k=1}^N \mu_k a_{Q_k}\|_{L^1(\mathbb{R}^d)} + \frac{2C}{\varepsilon} \sum_{k=N}^{\infty} |\mu_k|$$

(5)

and so, if choose $N \in \mathbb{N}$ such that

$$\bullet \|f - \sum_{k=1}^N \mu_k \alpha_{Q_k}\|_{L^1(\mathbb{R}^d)} \leq \frac{\varepsilon^2}{4C},$$

$$\bullet \sum_{k=N}^{\infty} |\mu_k| \leq \frac{\varepsilon^2}{4C},$$

we get $|\{x \in \mathbb{R}^d : |T(f)(x) - \sum_{k=1}^{\infty} \mu_k T(\alpha_{Q_k})(x)| > \varepsilon\}| < \varepsilon$ and this implies (3).

Note that (2) and (3) imply that $T(f) \in L^1(\mathbb{R}^d)$. ■

We have thus determined a subspace X of $L^1(\mathbb{R}^d)$ such that $T(f) \in L^1(\mathbb{R}^d)$ whenever $f \in X$. One is naturally led to ask the following informal question:

Q.: 'Is X in some sense optimal?'

A very precise answer can be given to the above informal question. More specifically, one can show that:

' $f \in X$ if, and only if, $R_j(f) \in L^1(\mathbb{R}^d)$ for all $j=1, \dots, d$, where R_j is the j -th Riesz transform on $\mathbb{R}^d$.

In particular, in the one-dimensional case one has $f \in X$ if, and only if, $H(f) \in L^1(\mathbb{R})$ with H being the Hilbert transform.'

Therefore, X is best possible because if $f \notin X$ then there exists a translation-invariant CZO T such that $T(f) \notin L^1(\mathbb{R}^d)$.

The story continues: ' $f \in X$ if, and only if, $f \in H^1(\mathbb{R}^d)$ where $H^1(\mathbb{R}^d)$ the real (variable) Hardy space on $\mathbb{R}^d$, which we define below.'

Definition 6. The real (variable) Hardy space $H^1(\mathbb{R}^d)$ is defined the class of all $f \in L^1(\mathbb{R}^d)$ such that $f^* \in L^1(\mathbb{R}^d)$, where

$$f^*(x) := \sup_{|x-t| < y} |(f * P_t)(y)| \quad (x \in \mathbb{R}^d)$$

$$\text{and } P_t(x) := \frac{\Gamma(d+1/2)}{\pi^{d+1/2}} \cdot \frac{t}{(t^2 + |x|^2)^{(d+1)/2}}$$

is the Poisson kernel on $\mathbb{R}^d$.

If $f \in H^1(\mathbb{R}^d)$, we set

$$\|f\|_{H^1(\mathbb{R}^d)} := \|f^*\|_{L^1(\mathbb{R}^d)}.$$

- The function f^* is called the non-tangential maximal function of $f \in L^1(\mathbb{R}^d)$.

Therefore, $H^1(\mathbb{R}^d)$ has the following properties:

- For any translation-invariant CZO T one has $T(f) \in L^1(\mathbb{R}^d)$ for all $f \in H^1(\mathbb{R}^d)$
- If $f \notin H^1(\mathbb{R}^d)$ there exists a $j \in \{1, \dots, d\}$ such that $R_j(f) \notin L^1(\mathbb{R}^d)$. Hence, if $f \notin H^1(\mathbb{R}^d)$ then there exists a translation-invariant CZO T such that $T(f) \notin L^1(\mathbb{R}^d)$.

Moreover, we have:

- ' $f \in H^1(\mathbb{R}^d)$ if, and only if, $S(f) \in L^1(\mathbb{R}^d)$, where $S(f)$ is the smooth Littlewood-Paley square function of f treated in the previous lectures.'

We can summarise the previous facts in the following theorem

Theorem 7. Let f be a function in $L^1(\mathbb{R}^d)$.

The following are equivalent:

- (a) $f \in H^1(\mathbb{R}^d)$,
- (b) there exists a sequence of complex numbers $\{\mu_k\}_{k \in \mathbb{N}} \in \ell^1(\mathbb{N})$ and a sequence of atoms $\{a_{Q_k}\}_{k \in \mathbb{N}}$ (see definition 1) such that
$$f = \sum_k \mu_k a_{Q_k} \quad \text{in } H^1(\mathbb{R}^d).$$
- (c) $R_j(f) \in L^1(\mathbb{R}^d)$ for all $j \in \{1, \dots, d\}$,
- (d) $S(f) \in L^1(\mathbb{R}^d)$.

Remark. The equivalence:

- (a) $\Leftrightarrow$ (b) is usually refer to as the 'characterisation of $H^1(\mathbb{R}^d)$ in terms of atoms' and is due to R. R. Coifman (1974) for $d=1$ and due to R. H. Latter (1978) for $d > 1$.
- (a) $\Leftrightarrow$ (d) is usually refer to as the 'Littlewood-Paley characterisation of $H^1(\mathbb{R}^d)$ ' and is due to E. M. Stein (1966).

We conclude this lecture with the following result of C. Fefferman and E.M. Stein (1972).

Theorem 8. The dual of $H^1(\mathbb{R}^d)$ can be identified with $BMO(\mathbb{R}^d)$.

In Theorem 8, $BMO(\mathbb{R}^d)$ denotes the space of functions that are of bounded mean oscillation on $\mathbb{R}^d$. We will not prove Theorems 7 and 8 in the lectures, but we will establish several interesting results for $BMO(\mathbb{R}^d) \cong (H^1(\mathbb{R}^d))^*$ in the forthcoming lectures

The notes of this lecture are based on the following references:

- [1] Duoandikoetxea, Javier. Fourier Analysis. Translated and revised from the 1995 Spanish original by David Cruz-Uribe. Graduate Studies in Mathematics, 29. American Mathematical Society, Providence, RI, 2001, xviii + 222 pp.
- [2] Grafakos, Loukas. Modern Fourier analysis. Third Edition. Graduate Texts in Mathematics, 250. Springer, New York, 2014, xvi + 624 pp.

For the results mentioned in the notes, see [1], [2].

